

Why Decoherence Cannot Replace Measurement

Outcome-Resolved Dynamics Without a Fundamental Measurement Postulate in Modal Triplet Theory

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Abstract

Measurement is not taken to be a fundamental act, an observer intervention, or a separate kind of physics. It is an ordinary system–apparatus–environment process described with a retained physical record. If that record is discarded, the same coupling is represented only by a nonselective channel. Decoherence can suppress interference and stabilize preferred records, but the record-discarded channel does not contain an outcome label or a conditional state update. We make this data distinction exact in finite-dimensional quantum theory and then state its consequence for Modal Triplet Theory (MTT). For a pointer resolution $\{P_a\}$, the dephasing channel $\Delta_P(\rho) = \sum_a P_a \rho P_a$ is completely positive and trace preserving, removes every off-diagonal pointer block, and preserves all pointer populations. We prove that the same nonselective channel is compatible with distinct outcome instruments having different probability laws. Hence the channel alone specifies neither records nor their probabilities.

An MTT basin boundary or chart exit has the same logical limitation: it marks the failure of a supplied effective chart, but it does not choose a successor basin. A complete outcome-resolved model must additionally provide an outcome-indexed completely positive instrument or a normalized selection-completion kernel, together with its source and domain. This yields a rigorous conditional bridge between record-discarded intra-basin suppression and record-resolved inter-basin continuation. Both are ordinary physical dynamics; the distinction concerns which data the effective description retains. The current canonical q79 binary recorder supplies an exact stopped-output law on its declared finite domain, but general apparatus contexts and objective single-history actualization remain open. The paper therefore retains the decoherence-versus-selection distinction while withdrawing any inference of capture, Born probabilities, irreversibility, or objective outcomes from projection, noninjectivity, admissibility loss, or chart exit alone.

Contents

1	Scope and Claim Tier	2
2	Channels, Pointer Blocks, and Decoherence	3
3	Outcome-Resolved Processes Require an Instrument	3
4	The Missing Selection-Completion Map	4
5	A Correct Conditional MTT Bridge	5
6	What Current MTT Results Supply	6

7	Relation to Standard Outcome-Resolved Dynamics	6
7.1	Decoherence and pointer stability	6
7.2	Instruments and trajectories	7
7.3	Redundant records	7
7.4	Collapse models	7
8	Consequences and Non-Consequences	7
8.1	Born weights	7
8.2	Irreversibility	7
8.3	Undecidability	7
8.4	Single outcomes	8
9	Outcome-Resolved Dynamics Promotion Contract	8
10	Version Delta	8
11	Conclusion	9

1 Scope and Claim Tier

Throughout this paper, “measurement” names no privileged physical process. It means an ordinary interaction whose macroscopic record is included in the description. The title’s phrase “cannot replace” is therefore a statement about information: a record-discarded marginal cannot provide the outcome-resolved data that were discarded.

Environment-induced decoherence is a dynamical account of interference suppression and pointer stability [2, 3]. The definite-outcome problem asks for something else: why an individual run has a recorded outcome, what its conditional post-measurement state is, and what law governs the alternatives. Conflating these questions obscures both the success of decoherence and the data still required for measurement.

This revision separates four claim tiers.

1. **Established measurement mathematics.** Channels describe nonselective dynamics; instruments describe outcome probabilities and conditional updates [1].
2. **Exact result of this paper.** A dephasing channel does not determine an instrument, and loss of an effective chart does not determine a continuation kernel.
3. **Conditional MTT bridge.** If a selected MTT descent emits basins, basin-preserving channels, and an inter-basin completion instrument, then record-discarded and record-resolved descriptions can be represented as intra- and inter-basin operations of the same physical system.
4. **Current selected finite-domain result.** The canonical q79 binary one-anchor recorder emits an operational stopped-output measure and second-moment capture descent on its declared domain [6]. This is not yet a universal apparatus theorem or an objective single-history theorem.

No result below derives a Born source, physical time, entropy arrow, or objective actualization from projection alone. The analysis is finite-dimensional so that every map and norm is elementary. The distinction extends to normal instruments on operator algebras, but that extension is not needed here.

2 Channels, Pointer Blocks, and Decoherence

Let $\mathcal{H}$ be a finite-dimensional Hilbert space and let

$$\mathcal{D}(\mathcal{H}) = \{\rho \geq 0 : \text{Tr } \rho = 1\}.$$

A finite pointer resolution is a family of mutually orthogonal projections $P = \{P_a\}_{a \in A}$ satisfying $\sum_a P_a = I$.

Definition 2.1 (Nonselective dephasing channel). The P -dephasing channel is

$$\Delta_P(\rho) = \sum_{a \in A} P_a \rho P_a. \quad (2.1)$$

Its pointer-coherence functional is

$$C_P(\rho)^2 = \sum_{\substack{a, b \in A \\ a \neq b}} \|P_a \rho P_b\|_2^2, \quad (2.2)$$

where $\|X\|_2^2 = \text{Tr}(X^\dagger X)$.

Proposition 2.2 (What dephasing does). *The map Δ_P is completely positive and trace preserving. For every $\rho \in \mathcal{D}(\mathcal{H})$,*

$$C_P(\Delta_P(\rho)) = 0, \quad \text{Tr}(P_a \Delta_P(\rho)) = \text{Tr}(P_a \rho)$$

for all $a \in A$.

Proof. Equation (2.1) is a Kraus representation with Kraus operators P_a , and $\sum_a P_a^\dagger P_a = I$. Orthogonality gives $P_a \Delta_P(\rho) P_b = 0$ for $a \neq b$, proving the first identity. Cyclicity of the trace and $P_a P_b = \delta_{ab} P_a$ give the second. $\square$

More general decoherence dynamics may approach Δ_P only asymptotically, may include dissipation, and may have an approximate pointer resolution. None of those refinements changes the type of the object: discarding the environmental record produces a nonselective quantum channel. It produces a density operator, not a classical outcome label.

Example 2.3 (Suppression without an outcome). For $\mathcal{H} = \mathbb{C}^2$, let $P_0 = |0\rangle\langle 0|$, $P_1 = |1\rangle\langle 1|$, and $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Then

$$\Delta_P(|+\rangle\langle +|) = \frac{1}{2}P_0 + \frac{1}{2}P_1.$$

The off-diagonal blocks vanish, but the output is not either conditional state P_0 or P_1 , and no recorded value 0 or 1 appears in the channel output.

3 Outcome-Resolved Processes Require an Instrument

An ordinary physical coupling can be represented at two resolutions. If its classical record is discarded, it gives a nonselective channel. If the record is retained, it gives an outcome-resolved instrument.

Definition 3.1 (Quantum instrument). A finite quantum instrument is a family $\{\mathcal{I}_a\}_{a \in A}$ of completely positive, trace-nonincreasing maps such that

$$\mathcal{I}_{\text{ns}} = \sum_{a \in A} \mathcal{I}_a$$

is trace preserving. For input ρ it defines

$$p(a \mid \rho) = \text{Tr } \mathcal{I}_a(\rho), \quad \rho_a = \frac{\mathcal{I}_a(\rho)}{p(a \mid \rho)} \quad \text{when } p(a \mid \rho) > 0. \quad (3.1)$$

The instrument contains three pieces absent from the record-discarded channel: an outcome index, its probability, and its conditional update. This does not introduce a fundamental measurement postulate. It is the outcome-resolved bookkeeping of the same kind of physical interaction. If one additionally claims that exactly one ontic history is actualized, that stronger ontology must identify what constitutes the realized event; it is not required for the ordinary operational meaning of a detector record.

Theorem 3.2 (A nonselective channel does not select an instrument). *The dephasing channel Δ_P does not determine outcome probabilities or conditional states. In particular, for the two-outcome qubit resolution of Example 2.3, the Lüders instrument*

$$\mathcal{I}_0(\rho) = P_0 \rho P_0, \quad \mathcal{I}_1(\rho) = P_1 \rho P_1$$

and, for every $q \in [0, 1]$, the instrument

$$\mathcal{J}_0^{(q)}(\rho) = q \Delta_P(\rho), \quad \mathcal{J}_1^{(q)}(\rho) = (1 - q) \Delta_P(\rho)$$

have the same nonselective channel Δ_P . On $|+\rangle\langle+|$ their outcome laws are respectively $(1/2, 1/2)$ and $(q, 1 - q)$.

Proof. Each displayed map is completely positive and trace nonincreasing. Both families sum to Δ_P . Their probabilities follow from Equation (3.1). Choosing $q \neq 1/2$ proves that the same nonselective channel is compatible with different outcome laws. Their conditional states also differ: the Lüders instrument gives P_a , whereas $\mathcal{J}_a^{(q)}$ gives the same dephased state for either nonzero outcome. $\square$

Corollary 3.3 (Decoherence non-selection). *No outcome law, conditional update, or unique realized outcome can be inferred from a decoherence channel without additional outcome-resolved data.*

The artificial family $\mathcal{J}^{(q)}$ is not proposed as a physical detector. Its purpose is logical: after the record is discarded, the channel does not encode which outcome-resolved process produced it. A source theorem must select the detector coupling, readout algebra, and instrument independently of the observed target probabilities.

4 The Missing Selection-Completion Map

The basin language used in MTT can be made precise without turning chart failure into an outcome theorem. Let $U \subseteq \mathcal{D}(\mathcal{H})$ be the domain of an effective chart and let $\{\mathcal{B}_a\}_{a \in A}$ be declared measurable successor regions.

Definition 4.1 (Selection-completion kernel). For a boundary state $\eta \in \partial U$, a selection-completion kernel is a probability kernel

$$K_{\partial}(a, d\sigma \mid \eta)$$

on $A \times \mathcal{D}(\mathcal{H})$ satisfying

$$\sum_{a \in A} \int_{\mathcal{D}(\mathcal{H})} K_{\partial}(a, d\sigma \mid \eta) = 1, \quad (4.1)$$

$$K_{\partial}(a, \mathcal{D}(\mathcal{H}) \setminus \mathcal{B}_a \mid \eta) = 0. \quad (4.2)$$

Its nonselective continuation is the barycenter

$$T_K(\eta) = \sum_{a \in A} \int_{\mathcal{D}(\mathcal{H})} \sigma K_{\partial}(a, d\sigma \mid \eta). \quad (4.3)$$

A deterministic continuation is the special case in which the kernel is a point mass. An affine completely positive realization is more strongly encoded by an instrument. Definition 4.1 is deliberately minimal: it completes the effective mathematical description; it does not add a special physical act. It exposes exactly the data that the phrase “capture into a basin” must supply.

Proposition 4.2 (Chart exit does not determine continuation). *The statement that a trajectory reaches ∂U does not determine K_{∂} . If two successor regions $\mathcal{B}_0, \mathcal{B}_1$ contain states σ_0, σ_1 , then both*

$$K_{\partial}^{(0)}(a, d\sigma \mid \eta) = \delta_{a0} \delta_{\sigma_0}(d\sigma)$$

and

$$K_{\partial}^{(1)}(a, d\sigma \mid \eta) = \delta_{a1} \delta_{\sigma_1}(d\sigma)$$

satisfy Equations (4.1) and (4.2), while producing different outcomes and continuations.

Proof. Both kernels are normalized point masses supported in their declared successor regions. Their outcome marginals and barycenters are different. Boundary membership supplies no condition that distinguishes them. $\square$

Thus admissibility loss has a precise, limited meaning: the current effective chart no longer controls the continuation. It does not imply that projection “enforces” one particular capture. A selected action, detector coupling, boundary condition, instrument, or kernel must do that work.

5 A Correct Conditional MTT Bridge

The useful core of the original paper can now be stated without overclaiming.

Definition 5.1 (Typed basin model). A typed basin model consists of:

- (i) a state space and declared basin regions $\mathcal{B}_a$;
- (ii) within-basin channels $\mathcal{D}_a$ satisfying $\mathcal{D}_a(\mathcal{B}_a) \subseteq \mathcal{B}_a$ on their stated domains;
- (iii) a pointer or record algebra;
- (iv) a selection-completion instrument or kernel at every claimed inter-basin event; and
- (v) a source certificate identifying how these objects descend from the same selected MTT data.

Theorem 5.2 (Conditional record-discarded–record-resolved bridge). *In a typed basin model, a channel $\mathcal{D}_a$ that suppresses off-diagonal pointer blocks while preserving $\mathcal{B}_a$ is an intra-basin operation. A completion kernel with support in $\mathcal{B}_b$, $b \neq a$, is an inter-basin operation. These may be two resolutions of one ordinary physical system–apparatus process. The record-discarded channel does not determine the record-resolved kernel.*

Proof. The first two statements are immediate from the declared domains and support conditions. The last follows from Theorem 3.2 at channel level and Proposition 4.2 at chart-boundary level. $\square$

This theorem retains the intended distinction:

record discarded $\longleftrightarrow$ intra-basin suppression and record stabilization,
 record retained $\longleftrightarrow$ an outcome-indexed inter-basin instrument or kernel.

The arrow is a typed representation, not an ontological division into “measurement physics” and ordinary physics, and not a proof that both sides have already been emitted by one universal MTT source.

6 What Current MTT Results Supply

Several earlier formulations attributed too much to noninvertible projection. A noninjective map can identify microscopic states without selecting a time arrow, an entropy law, a probability measure, or a representative of each fiber. The corrected MTT shadow-bridge analysis states exact descent and restricted-recovery criteria and treats measurement instruments as selected updates, not inverse maps [4].

There is nevertheless a concrete positive result. On the canonical q79 binary one-anchor Fock recorder, the selected normal state on the commuting nondemolition output algebra induces a stopped output law, and second-moment capture descent is exact on the declared finite-symbol domain [6]. No observed probability or fitted stochastic parameter is added on that domain.

That theorem closes an operational finite-domain row; it does not prove:

- the corresponding construction for every allowed apparatus context;
- finite-bandwidth and non-Markov corrections;
- a pre-quantum probability semantics, if one is demanded;
- objective actualization of one ontic history; or
- that every admissibility boundary carries the same completion law.

Those statements require their own same-source theorems.

7 Relation to Standard Outcome-Resolved Dynamics

7.1 Decoherence and pointer stability

Standard decoherence theory derives suppression of interference in reduced states through system–environment entanglement and explains the dynamical selection and stability of preferred pointer structures [2, 3]. Proposition 2.2 is the finite idealization of this nonselective effect. It agrees with, rather than replaces, the standard account.

7.2 Instruments and trajectories

The Davies–Lewis instrument formalism associates classical outcome statistics and conditional quantum operations with a measurement [1]. Quantum-trajectory descriptions add a monitored record and an unravelling of a nonselective master equation. Different unravellings can represent the same ensemble channel, so the master equation alone does not select a unique record process. This is not evidence for a special measurement force. It says only that MTT must emit the ordinary detector coupling, physical readout, and outcome-resolved instrument, not merely reproduce their record-discarded reduced channel.

7.3 Redundant records

Environmental redundancy can explain why information in a pointer observable is robustly accessible. Redundancy is compatible with the distinction made here: it concerns the proliferation of record information, whereas Theorem 3.2 concerns the missing outcome-resolved law when only the nonselective channel is supplied. No claim that Darwinism presupposes a particular MTT selection mechanism is needed.

7.4 Collapse models

Objective-collapse models append nonlinear or stochastic state dynamics that does more than suppress ensemble coherences. Whether an MTT limit reproduces one such model is a separate source-and-error problem. Even an exact dephasing semigroup can possess a random-unitary unravelling and therefore does not by itself prove objective collapse [5].

8 Consequences and Non-Consequences

8.1 Born weights

If the Lüders instrument is supplied, then

$$p(a \mid \rho) = \text{Tr}(P_a \rho)$$

follows immediately. This is an evaluation of a selected instrument, not a derivation of that instrument from decoherence. Basin sizes can represent arbitrary finite probability vectors, so matching target weights by basin volume is not predictive unless the measure and partition were independently selected [4].

8.2 Irreversibility

Reduced decoherence may be effectively irreversible because environmental records are inaccessible, and an instrument may produce persistent records. Neither fact follows from noninjectivity alone. An arrow of time requires a declared state, boundary condition, semigroup, record algebra, or other asymmetric source.

8.3 Undecidability

No algorithmic undecidability theorem follows from the existence of basin boundaries. Such a theorem requires a computably specified MTT decision problem and an explicit reduction from a known undecidable problem. The earlier claim that quantum probability arises because selection is undecidable is therefore withdrawn.

8.4 Single outcomes

An operational instrument predicts the distribution and conditional state of records. It does not by itself settle whether one ontic history is fundamental, emergent, or interpretation-relative. Any stronger single-history claim must identify an actualization object and prove its relation to the instrument.

9 Outcome-Resolved Dynamics Promotion Contract

A universal MTT detector-and-record theorem must emit, from one selected source:

1. the physical state and observable algebras;
2. the pointer or record algebra and apparatus coupling;
3. the nonselective channel with its approximation domain and errors;
4. the outcome instrument or completion kernel;
5. normalization, positivity, and coarse-graining consistency;
6. the probability source without observed target weights as inputs;
7. stable record formation;
8. any claimed objective actualization rule; and
9. transport to general apparatus contexts, including finite-bandwidth and non-Markov regimes.

The canonical q79 recorder supplies a nontrivial subset of this contract on one finite domain. The remaining rows should be attacked as source and transport theorems, not filled by interpreting chart exit as selection.

10 Version Delta

Relative to version 1, this revision:

- retains the distinction between intra-basin suppression and inter-basin selection;
- states explicitly that measurement is ordinary record-forming physics, not a fundamental act or observer-dependent postulate;
- defines decoherence as a nonselective channel and proves its exact suppression properties;
- adds an explicit quantum instrument and a selection-completion kernel;
- proves that one nonselective channel supports different instruments and probability laws;
- proves that chart exit does not select a successor basin or kernel;
- replaces asserted projection-enforced capture by a conditional typed basin theorem;
- records the exact finite-domain q79 recorder result at its current tier;
- withdraws Born, irreversibility, undecidability, and objective-outcome inferences from projection or admissibility loss alone; and
- adds a testable promotion contract for future universal measurement claims.

11 Conclusion

Measurement is ordinary physics. Nothing in this paper gives it a fundamental status, a privileged observer, or a separate force. The title means only that a record-discarded description cannot supply the outcome-resolved data it omits. A nonselective channel can erase off-diagonal pointer blocks and stabilize records while containing no outcome index. Retaining the record requires an instrument or completion kernel that supplies outcome probabilities and conditional updates. The same channel can be associated with different outcome-resolved instruments, and the same chart exit can be continued by different kernels.

This does not invalidate the MTT basin picture. It makes the picture precise. Intra-basin suppression and inter-basin record formation may be stages or resolutions of one physical process once both are emitted from selected source data. MTT already has an exact operational account on one canonical finite q79 recorder domain. Extending that result to general apparatus contexts and controlled memory effects is the operational frontier. Objective single-history actualization is a separate, stronger question, not part of what makes an ordinary interaction a measurement.

References

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